Mangirish Samant

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PROFESSIONAL SUMMARY

Final-year Computer Engineering student building **end-to-end data engineering, analytics and ML projects** in Python and SQL: a **PySpark / Spark SQL** pipeline over 76M government records, an **NLP** search and classification app on parliamentary questions, and tuned gradient-boosting models deployed with **FastAPI + Docker**. Hands-on with **Pandas, Scikit-learn, LightGBM, XGBoost, SHAP, Flask and Streamlit**. Seeking **Data Analyst / Data Scientist** roles.

EDUCATION

MCT's Rajiv Gandhi Institute of Technology, Mumbai *2023 – 2027*
Bachelor of Engineering (B.E.) in Computer Engineering **CGPA: 8.76 / 10.0**
HSC (12th): 84.50% • **SSC (10th):** 86.20%

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript

Big Data: PySpark, Spark SQL, Parquet, Medallion Architecture, Streaming Algorithms (Flajolet–Martin)

NLP: Text Preprocessing, TF-IDF, LSA, Topic Modelling (NMF), Named Entity Recognition, Text Classification, Extractive Summarisation (TextRank)

Data Analysis: Pandas, NumPy, Polars, DuckDB, Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Statistical Testing

Machine Learning: Scikit-learn, LightGBM, XGBoost, CatBoost, Optuna, SHAP, Classification, Regression, Clustering (K-Means), Imbalanced Learning, Hyperparameter Tuning

Visualization & BI: Power BI, Streamlit, Plotly, Matplotlib, Seaborn, Excel

Deployment & Tools: FastAPI, Flask, REST APIs, Docker, MySQL, Git, GitHub, Jupyter Notebook, Linux

PROJECTS

MandiPulse – India Crop Price Analytics (Big Data) [GitHub] *PySpark, Spark SQL, Parquet, Streamlit, Plotly*

- Built a PySpark pipeline over **76M daily price records** (6.5 GB, 26 years) from the Government of India's AGMARKNET mandi data, cleaning them to **74.2M valid rows** across 7,278 mandis and 397 commodities, stored as year-partitioned Parquet (bronze/silver/gold layers).
- Wrote **Spark SQL** (CTEs, window functions, LAG, RANK, rolling 12-month z-scores, approximate medians) to build 9 summary tables on price volatility, seasonality, price shocks and state-vs-national price gaps.
- Implemented the **Flajolet–Martin** streaming algorithm in Spark SQL to estimate distinct mandis with 64-byte sketches, validated against exact counts and HyperLogLog; presented everything in a 5-page **Streamlit + Plotly** dashboard.

SansadLens – Lok Sabha Questions NLP Search & Analytics [GitHub] *Python, Scikit-learn, NLP, Flask*

- Built a Flask web app that searches **4,750 parliamentary questions** MPs asked ministers (Winter Session 2021), using hybrid **TF-IDF + LSA** semantic search with State-aware ranking.
- Trained a ministry classifier (Logistic Regression) across **51 ministries**: **78.1% accuracy, 91.0% top-3, 0.756 macro F1** on 948 held-out questions vs a 6% baseline, after removing a label-leaking opening phrase.
- Added NMF topic modelling (14 themes), rule-based NER for States, constituencies and schemes, and **TextRank + MMR** summaries.

Credit Risk & Loan Default Prediction System [GitHub] *Python, SQL, LightGBM, SHAP, FastAPI, Docker*

- Built a loan default model on the Home Credit dataset, joining **7 relational tables** and engineering repayment-behaviour features with **SQL (DuckDB)** and Featuretools; selected the **top 15** features via LightGBM importance and statistical tests.
- Tuned LightGBM and Decision Tree models with **Optuna** and tuned the decision threshold for a highly imbalanced target (~8% defaulters), raising precision from **0.17 to 0.21 at 50% recall**; explained drivers with **SHAP** and served the model via a **FastAPI** endpoint in **Docker**.

Customer Cross-Sell Prediction (Vehicle Insurance) [GitHub] *Python, Scikit-learn, LightGBM, XGBoost, CatBoost*

- Predicted which of **381K** insurance customers would buy vehicle insurance (Analytics Vidhya JanataHack); EDA showed customers with past vehicle damage convert at **23.8% vs 0.5%** and already-insured ones almost never (**0.1%**), so campaigns can skip them.
- Compared Random Forest, XGBoost, CatBoost, LightGBM and a neural network with stratified K-fold; LightGBM scored **0.855 ROC-AUC** on validation and **0.863** on the private leaderboard.

Other work: Bulldozer price regression on 412K sales (R^2 0.88); heart disease classifier (88.5% accuracy); Android quiz app (Java); news aggregator web app (JavaScript, MySQL).

CERTIFICATIONS & LEADERSHIP

- **Certifications:** Google Cloud Engineering (2026) • Anthropic AI Fluency • Elements of AI (Univ. of Helsinki)
- **Web Secretary – DMX Team (2026–27):** Managed web development initiatives and platform maintenance.